Final Year Projects

Project Presentation

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Project 1

An Information-Theoretic Inequality for the Information Bottleneck

First Project Overview

Background

- A new information-theoretic inequality involving three Boolean random variables has been conjectured.
- It extends the well-known **Mrs. Gerber's Lemma** from classical information theory.
- It has deep connections to the **Information Bottleneck** principle in machine learning.

Project Goals

- Explore optimization-based approaches to investigate and verify the conjecture.
- Develop potential generalizations and related extensions of the result.

The Conjecture

Conjecture

Let $p_{X,Y}$ be an arbitrary distribution on non-constant binary random variables X and Y.

Let ρ be defined as:

$$\rho = \frac{\sqrt{p_{Y|X}(1|0)p_{Y|X}(0|1)}}{\sqrt{p_{Y|X}(1|0)p_{Y|X}(0|1)} + \sqrt{p_{Y|X}(0|0)p_{Y|X}(1|1)}}$$

Then, for any auxiliary random variable W satisfying the Markov chain $W \to X \to Y$, we have

$$I(W;Y) \leq 1 - H_2\big(\rho \star H_2^{-1}(1 - I(W;X))\big),$$

where $a \star b = a(1-b) + (1-a)b$ and $H_2(\cdot)$ is the binary entropy function.

Project 2

f-Divergence Information Inequalities

Background & Foundational Concepts

- **Shannon's Mutual Information:** A fundamental measure used to quantify the dependency and shared information between two random variables.
- **Broad Applications:** Serves as a cornerstone for communication systems, machine learning, and statistical inference.
- **Generalization via f-Divergence:** Provides a broader, generalized framework for measuring the divergence between probability distributions.
- **Core Theoretical Topics:** As part of the project foundation, we will engage with key concepts including:
  - Entropy
  - Kullback-Leibler (KL) Divergence
  - Mathematical properties of f-divergences

Project Objectives & Methodology

Primary Objective

To identify a comprehensive list of feasible information inequalities that are satisfied by information measures defined via f-divergence.

- **Methodology (Simulation-Based):**
  - Conduct extensive numerical simulations to explore and verify potential inequalities.
  - Computationally map out the boundaries of these information measures.
- **Expected Outcomes:**
  - Deepen our understanding of the theoretical relationships between different information-theoretic measures.
  - Uncover the mathematical constraints that govern generalized mutual information.